

*MA.7.DP.1.4***Benchmark**

MA.7.DP.1.4 Use proportional reasoning to construct, display and interpret data in circle graphs.

Benchmark Clarifications:

Clarification 1: Data is limited to no more than 6 categories.

Connecting Benchmarks/Horizontal Alignment

- MA.7.NSO.1.2
- MA.7.GR.1.4
- MA.7.AR.3.1

Terms from the K-12 Glossary

- Circle Graph
- Data
- Proportional Relationships

Vertical Alignment**Previous Benchmarks**

- MA.6.AR.3.4

Next Benchmarks

- MA.912.DP.1.1, MA.912.DP.1.2

Purpose and Instructional Strategies

In grade 6, students worked with solving problems using ratio relationships and their percentages. In grade 7, students apply their knowledge of ratios to solve problems involving proportions, including using proportional reasoning to construct, display and interpret categorical data in circle graphs. In high school, students will select an appropriate method to represent numerical or categorical data and univariate or bivariate.

- Circle graphs can be used to show how categories represent part of a whole, or compositions. Totals are represented as percentages that equal 100%, which illustrates the percentage breakdown of items and visually represents a comparison.
- Students should be able to identify strengths and limitations in showcasing data within a circle graph.

Strengths	Limitations
• Compares parts to a whole and to each other • Each response is shown as a percent • Size of sectors compare results	• Graph does not show total populations • Totals require additional information that is often not represented in the graph • Difficult to draw by hand, need tools or software for accurate depictions • Circle graphs are not effective, however, when there are too many categories.

- Instruction begins with data sets out of 100 to allow for easier calculations of percentages.

- Instruction should include opportunities for students to brainstorm how they might split up their circle into the needed percentages (*MTR.5.1*).
 - For example, students can slice a circle into 4 equal parts to show students the 4 right angles at the center which total 360°. Then use proportional relationships to determine the central angle sizes needed based on the percentage size of each “slice” of the circle.
- Students should collect their own data with which to create a circle graph (*MTR.7.1*).
 - For example, have students count colored candy/snacks or survey other students in the room about their favorite color, favorite sport or favorite genre of music/movies.
- Use protractors or online software to assist in creating circle graphs accurately.

Common Misconceptions or Errors

- Students may incorrectly use the percent of a category for the central angle degrees instead of finding the degrees by using a proportion.
- Students may incorrectly round or make other errors in calculations that will lead to the circle graph sections not totaling 100%.
- Students may incorrectly use a protractor. To address this issue, provide students with a circle and allow them to measure sections and then find the percent.

Strategies to Support Tiered Instruction

- Teacher models several examples to work through with students, showing how to set up the proportion to find the central angle degrees, referencing patterns for students to discuss.
 - For example, if students need to determine the angle measure that corresponds to 21%, the proportion below can be used.

$$\frac{21}{100} = \frac{x}{360}$$

- Teacher models and works through several problems while discussing how to properly round when having to total to 100%, reinforcing to students to work through the problems carefully as to not make computation errors.
- Teacher models using computer-based software to create circle graphs to verify appropriate rounding.
- Teacher provides students with fill in the blank examples working from percent of a category using proportions to find the central angle degrees.
- Teacher provides several completed examples of problems where rounding was needed for students to reference while working through multiple problems together.
- Teacher models using fraction circle manipulatives to support converting fractions to percentages.

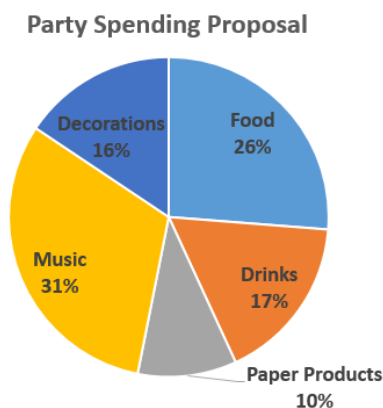
Instructional Tasks

Instructional Task 1 (MTR.4.1, MTR.7.1)

A group of friends has been given \$800 to host a party. They must decide how much money will be spent on food, drinks, paper products, music and decorations.

Part A. As a group, develop two options for the friends to choose from regarding how to spend their money. Decide how much to spend in each area and create a circle graph for each option to represent your choices.

Part B. Mikel presented the circle graph below with his recommendations on how to spend the money. How much did he choose to spend on food and drinks? How much did he choose to spend on music?



Instructional Items

Instructional Item 1

Circle Point High School surveyed its students to determine which mode of transportation they use to get to and from school. Create and label a circle graph based on the results given below.

Mode of Transportation	Walk/Bicycle	Drive	Parent Pickup	Bus
Number of Students	96	224	86	694

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*